AI ENGINEERING / APPLIED AI

PROOF AI

04

PROPOSED RESEARCH / NOT YET VERIFIED

THE ENGINEERING QUESTION

How much incorrect automated acceptance can independent checks and abstention prevent without making automation impractical?

PROBLEM DEFINITION AND RESEARCH CONTRIBUTION

A verifiable AI workflow that separates probabilistic document interpretation from deterministic financial controls and explicit human review.

- This should solve a trust problem, not just showcase another extraction pipeline.
- The research value lies in what deterministic controls catch, what they miss and how review thresholds affect throughput.
- Use synthetic or openly licensed data only; never imply production compliance certification.

ARCHITECTURE AND IMPLEMENTATION SPECIFICATION

1 STAGE **Document input** → 2 STAGE **Model extraction** → 3 STAGE **Rule verification** → 4 STAGE **Approve / review**

- Compact local models or OCR engines feeding typed schemas, cross document matching and arithmetic validation.
- Approval workflow with provenance, persistent state, replay and explicit rule outcomes for each decision.
- Evaluation dataset versioning, error taxonomy, calibration plots and operational tradeoff reporting.

ENGINEERING DECISIONS TO MAKE EXPLICIT

The design review must record data contracts, state transitions, failure handling and why the selected baseline is defensible. Every subsystem should have an interface specification and a reproducible test route.

DESIGN BOUNDARIES AND ASSUMPTIONS

⚠ Rule consistency does not guarantee factual correctness.

⚠ Hybrid systems can still fail on coherent but wrong upstream interpretations.

ⓘ Scope is proposed unless explicitly identified as documented existing work. All measured claims require reproducible evidence.

PROOF AI / PROTOCOL / ACCEPTANCE / REVIEW

EXPERIMENTS, VERIFICATION & PUBLICATION

04

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EXPERIMENTAL METHODOLOGY

01 Build labelled synthetic documents with duplicates, missing fields, OCR noise, total conflicts and identity mismatches.

02 Implement model only, rules only and hybrid baselines with persistent provenance and review states.

03 Evaluate confidence calibration, abstention thresholds, failure categories and undetected but consistent errors.

MEASUREMENT AND ACCEPTANCE EVIDENCE

- False automatic accept rate vs human review volume
- Extraction precision/recall and confidence calibration
- Examples of errors that pass every deterministic rule

REQUIRED REPOSITORY ARTEFACTS

- Versioned evaluation dataset
- Decision replay and review interface
- Technical report with calibration and failure analysis

TECHNICAL REVIEW QUESTIONS

Which assumptions could invalidate the result? How do the baselines behave under the same conditions? What is the strongest negative case? Which variables, seeds, workloads, software versions and machine properties must be retained for independent reproduction?

VERIFICATION MAPPING

For each acceptance claim, record the test identifier, input configuration, observed result, pass/fail definition and an independent rerun procedure. Failed or inconclusive cases remain in the report rather than being silently excluded.

PUBLICATION DISCIPLINE

Release original source, configuration files, test harnesses, raw measurements and scripts where rights permit. Distinguish simulation outcomes from physical validation and completed achievements from planned evaluation. Results must not be inferred from targets.